Q1. DIRECTIONS *for questions 1 and 2:* Type in your answer in the input box provided below the question.

A starts a certain work and after he completes exactly half the work, he leaves, and B takes up the remaining work and it takes a total of 25 days for the work to be completed this way. If A and B working together can complete the work in 12 days, and A is slower than B, then find the number of days in which the work will be completed, if A alone first works on exactly one-fifth of the work, after which B alone completes the rest of the work.

Q2. DIRECTIONS *for questions 1 and 2:* Type in your answer in the input box provided below the question.

If the distance between the points at which the straight line $x = n$ $(n > 0)$ intersects the curves $f(x) = \log_9(x + 6)$ and $g(x) = \log_9(x + 2)$ is $\frac{1}{4}$, find the value of $\frac{n}{\sqrt{3}}$.

Q3. DIRECTIONS *for questions 3 to 6:* Select the correct alternative from the given choices.

If the area of a quadrant of a circle is 154 cm^2, what is its perimeter? (Assume $\pi = 22/7$)

- a) **22 cm** ✗ Your answer is incorrect
- b) **36 cm**
- c) **50 cm**
- d) **44 cm**

Q4. DIRECTIONS *for questions 3 to 6:* Select the correct alternative from the given choices.
There are five positive integers. When any four of these integers are considered and their average is added to the fifth integer, we get the following numbers: 41, 44, 50, 56 and 65. Which of the following gives the value of one of these five integers?

- a) **28**
- b) **44**
- c) **36**
- d) **18**

Q5. DIRECTIONS *for questions 3 to 6:* Select the correct alternative from the given choices.

Consider 20 infinite geometric progressions, whose first terms are 2, 3, 4, ……21 respectively, and common ratios are $\frac{1}{3}, \frac{1}{4}, \frac{1}{5} \cdots\cdots \frac{1}{22}$ respectively. If S_1, S_2 , S_3……S_{20} respectively denote the sums of these 20 geometric progressions, find $S_1 + S_2 + S_3 + \ldots\ldots + S_{20}$.

- a) **230**
- b) **320**
- c) **160**
- d) **250**

Q6. **DIRECTIONS** *for questions 3 to 6:* Select the correct alternative from the given choices.

The area of a rectangle is 675 sq.cm. If both the length and the breadth (in cm) of the rectangle are integers, how many such rectangles are there, for which at least one of the dimensions is not a multiple of 3 and at least one of the dimensions is not a multiple of 5?

- a) **2** ✓ Your answer is correct
- b) **3**
- c) **4**
- d) **5**

DIRECTIONS *for questions 7 and 8:* Answer the questions on the basis of the information given below.

A five-digit number '*abcde*' is such that it is divisible by 6 and $a < b < c < d < e$.

Q7. DIRECTIONS *for questions 7 and 8:* Select the correct alternative from the given choices.
If $e < 8$, then how many such five-digit numbers are possible?

a) **1**

b) **2**

c) **4**

d) **12**

DIRECTIONS *for questions 7 and 8:* Answer the questions on the basis of the information given below.

A five-digit number '*abcde*' is such that it is divisible by 6 and $a < b < c < d < e$.

Q8. **DIRECTIONS** *for questions 7 and 8:* Select the correct alternative from the given choices.
How many such five-digit numbers are possible in all?

a) **13**

b) **14**

c) **17**

d) **15**

Q9. DIRECTIONS *for question 9:* Type in your answer in the input box provided below the question.

$f(x)$ is a quadratic function, the roots of which are α and β. If $f(4) = 4f(1)$ and $\beta = 3$, find the value of 12α.

Q10. DIRECTIONS *for question 10:* Select the correct alternative from the given choices.

If $f(x)$ is defined as the greatest integer less than or equal to $\frac{x}{2}$, how many of the following statements are true?

I. $f(x)$ is half the least even integer greater than or equal to x.

II. $f(x)$ is the least integer greater than or equal to $\frac{x}{2} - 1$.

III. $f(x)$ is one less than half the least even integer greater than x.

- a) **0**
- b) **3**
- c) **2**
- d) **1**

Q11. DIRECTIONS *for questions 11 to 13:* Type in your answer in the input box provided below the question.

The vertical distance covered by a body falling freely under gravity is directly proportional to the square of the time for which it falls. If a body covers 300 m in the first 10 seconds of its free fall, what is the distance (in m) covered by the body in the next 10 seconds of its fall?

Q12. DIRECTIONS *for questions 11 to 13:* Type in your answer in the input box provided below the question.

What is the remainder when 7^{700} is divided by 100?

Q13. DIRECTIONS *for questions 11 to 13:* Type in your answer in the input box provided below the question.

If the lines $2x + 3y - 5 = 0$, $2x + 3y - 10 = 0$, $3x + 2y - 5 = 0$ and $3x + 2y - 10 = 0$ represent the sides of a parallelogram, then find the area (in sq. units) of the parallelogram.

Q14. DIRECTIONS *for question 14:* Select the correct alternative from the given choices.

If $-8 \le x \le -7$ and $-6 \le y \le 6$, where x and y are non-zero integers, which of the following must be true?

a) $-13 \le (x + y) \le 2$

b) $-42 \le xy \le 48$ ✗ Your answer is incorrect

c) $-8 \le \frac{x}{y} \le 8$

d) **More than one of the above**

Q15. DIRECTIONS *for question 15:* Type in your answer in the input box provided below the question.

If K is a natural number and $(K^2 - 10K + 21)(K^2 - 9K + 20) = 240$, find K.

Q16. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.

The ratio of milk and water in a mixture of the two is 3 : 1. First, the volume of the mixture is increased by 50% by adding water. Next, 25 litres of the mixture is replaced with water. If the final ratio of milk and water in the resultant mixture is 1 : 3, find the initial quantity of mixture present (in litres).

a) $16\frac{2}{3}$

b) $33\frac{1}{3}$ ✓Your answer is correct

c) **40**

d) **80**

Q17. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.

Everyday, Rishab's mother starts from home, drives to his school and reaches there at exactly 4:00 p.m., and immediately picks him up from there and drives him back home. One day, waiting for his mother, started walking towards home. He met his mother on the way and together they reached home $13\frac{1}{3}$ minutes earlier than usual. If the next day, he left school at 3:30 p many minutes earlier than usual did they reach home?

- a) **12**
- b) **8**
- c) $6\frac{2}{3}$
- d) $6\frac{1}{3}$

Q18. **DIRECTIONS** *for questions 16 to 20:* Select the correct alternative from the given choices.

In a class of 120 students, each student took three tests – Maths, English and Humanities. The number of students who passed in the three tests were 60, 90 and 80 respectively. If none of the students failed in all three subjects and the number of students who passed in at least one subject was 25% more than those who passed in at least two subjects, then how many students passed in all three subjects?

- a) **18**
- b) **10**
- c) **12**
- d) **14**

Q19. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.

If $\frac{7}{b}+\frac{8}{a}=-1$, where a and b are non-zero integers, for how many values of (a, b), is $(a + b)$ positive?

- a) **6**
- b) **2**
- c) **4**
- d) **8**

Q20. DIRECTIONS *for questions 16 to 20:* Select the correct alternative from the given choices.

If T = $2\cos^4\theta + \sin^2\theta + 3$, then the range of T is

a) $\left[\frac{1}{2}, 5\right]$

b) **[2, 3]**

c) $\left[\frac{31}{8}, 5\right]$

d) $\left[\frac{31}{7}, 4\right]$

Q21. DIRECTIONS *for question 21:* Type in your answer in the input box provided below the question.

The decimal equivalent of $(123.21)_4$ is

Q22. DIRECTIONS *for question 22:* Select the correct alternative from the given choices.

Raghu wants to construct a circular swimming pool in his plot which is in the shape of a trapezium. The lengths of the oblique sides of the plot are 26 m and 30 m. If the radius of the biggest circle that can be marked for the pool is 12 m, what must be the minimum area (in sq. m) of the plot?

- a) **336**
- b) **672**
- c) **728**
- d) **1344**